

Technology Stack

Date	5 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, JavaScript	Used to create a responsive and user friendly interface for entering weather parameters and displaying flood prediction results.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python 3, Flask	Flask handles HTTP requests, processes user input, loads the trained machine learning model and returns prediction results efficiently
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	SQLite (database.db)	A serverless, lightweight SQL database integrated directly into Python. It requires no separate setup making it efficient for this deployment. It persists two relational tables: users (storing credentials) and history (recording meteorological inputs, calculated

			probabilities and predictions linked to each user).
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Python Anywhere	Selected for simple and cost effective hosting of Python based web apps. The automated deployment script (deploy.py) uses the platform's REST API to upload files, generate a WSGI endpoint, install dependencies from requirements.txt and reload the server.
5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Git, Github	Used to track local file edits, manage the source repository history and coordinate code backups and transfers.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	PythonAnywhere REST API, Matplotlib & Seaborn (for generating analysis figures), NumPy, Pandas, VS Code	The PythonAnywhere REST API facilitates script-based automated deployment. Matplotlib and Seaborn generate the EDA charts (distributions, heatmaps, model comparisons and confusion matrices) that are served on the analysis page. Pandas & NumPy are used for data processing and analysis, VS Code is used to build the Flask application